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WEEK 7: Deliverable 1- Prompt Log PDF

Section 1:- 10 BI Prompts-

Sr No.	Task	Prompt	AI Tool Used	Output (Summary)
1	Data Cleaning Steps for a Messy Dataset	You are a Business Intelligence Analyst preparing the NYC Airbnb 2019 dataset for dashboard development. Recommend a complete data cleaning workflow, including handling missing values, duplicates, outliers, incorrect data types, inconsistent values, and feature engineering.	ChatGPT	Generated a structured workflow covering duplicate removal, missing value handling, data type correction, outlier detection, category standardization, feature engineering, and data validation before analysis.
2	Defining 3 KPIs for a Business	You are a BI Consultant working on the StaySense BI project. Using the NYC Airbnb 2019 dataset, identify three KPIs that should be monitored to evaluate listing performance and explain why each KPI is important.	ChatGPT	Suggested Average Listing Price, Number of Customer Reviews, and Room Type Distribution as the primary KPIs, with explanations of their business relevance.
3	Writing an Executive Summary from a Dashboard	You are a Senior BI Analyst preparing a report for executives. Based on a dashboard showing Airbnb prices, room types, review activity, and borough analysis, write an executive summary highlighting major findings, business impact, and recommendations.	ChatGPT	Produced a concise executive summary emphasizing premium pricing in Manhattan, customer engagement patterns, and strategic recommendations.
4	Generating SQL Queries from Natural Language	Generate an SQL query to calculate the average Airbnb listing price for each neighbourhood group using the NYC Airbnb dataset. Sort the results in descending order and use meaningful column aliases.	ChatGPT	Generated a correct SQL query using AVG(), GROUP BY, ORDER BY, and descriptive aliases for reporting.

5	Recommending a Chart Type	Recommend the most appropriate visualization to compare average Airbnb listing prices across neighbourhood groups. Explain why it is suitable and suggest one alternative with its limitation.	ChatGPT	Recommended a bar chart for category comparison and suggested a box plot as an alternative for distribution analysis.
6	Interpreting a Correlation	Explain the business meaning of a weak correlation between Airbnb listing price and number of customer reviews using simple, non-technical language.	ChatGPT	Explained that higher-priced listings do not necessarily receive more reviews and suggested that location and room type may influence customer engagement more strongly.
7	Detecting Data Quality Issues	Review the NYC Airbnb dataset and identify possible data quality issues that could affect BI reporting. Suggest practical solutions for each issue before analysis.	ChatGPT	Identified missing values, duplicate records, outliers, inconsistent categories, and incomplete information, along with appropriate cleaning techniques.
8	Writing a Slide Headline	A dashboard chart shows that Manhattan has the highest average Airbnb listing price. Generate five concise executive-level slide headlines communicating this finding.	ChatGPT	Produced several impactful presentation headlines highlighting Manhattan's premium pricing and market position.
9	Drafting a Business Recommendation	Based on the finding that Entire Home/Apartment listings receive the highest customer engagement, recommend three business actions that Airbnb hosts should implement.	ChatGPT	Recommended promoting Entire Home listings, adopting dynamic pricing, and improving listing visibility through targeted marketing.
10	Explaining a Technical Concept	Explain Business Intelligence to a non-technical Airbnb host. Describe how dashboards help improve listing performance using simple language without technical jargon.	ChatGPT	Explained Business Intelligence as a method of turning data into useful insights through dashboards that support better pricing and business decisions.

Section 2:- 3 Variation Comparison-

Variation	Prompt	Output (Summary)
Variation 1 – Minimal	Write an executive summary of this Airbnb dashboard.	The AI generated a short, generic summary describing the dashboard without specific insights or business recommendations.
Variation 2 – Context-Rich	You are a BI Analyst. The dashboard analyzes the NYC Airbnb 2019 dataset and includes average listing prices by borough, room type distribution, customer review activity, and price categories. Write a 150-word executive summary for the CEO highlighting the main findings and business implications.	The AI produced a more detailed summary describing Manhattan's higher prices, room type trends, and customer engagement while providing relevant business insights.
Variation 3 – Structured + Role	Act as a Senior Business Intelligence Consultant preparing a Board Report. Based on the StaySense BI dashboard, write a 150-word executive summary using three sections: **Key Findings**, **Business Impact**, and **Recommendations**. Use professional business language, avoid jargon, and mention only insights supported by the dashboard.	The AI generated a structured, executive-ready summary with clear sections, actionable recommendations, and findings directly supported by the dataset.

Comparison-

The structured role-based prompt (Variation 3) produced the highest-quality output because it clearly defined the audience, writing style, response format, and constraints. Compared to the minimal prompt, it generated more relevant, organized, and actionable insights. The context-rich prompt improved accuracy by providing dataset information, but the addition of role prompting and structured formatting made the final response significantly more professional and suitable for executive reporting. This comparison demonstrates that providing context, defining the audience, and specifying the desired output format greatly improve the quality of AI-generated business intelligence content.